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SUPERSEDING
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18 October 1963

MILITARY SPECIFICATION

THEODOLITE, SURVEYING: DIRECTIONAL, ASTRONOMICAL, 0.1 SECOND, WITH ACCESSORIES, TRIPOD, AND CASES

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a direction-type theodolite with accessories, tripod, and cases.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

FSC 6675

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

TT-V-121	- Varnish, Spar, Water-Resisting.

MILITARY

MIL-V-173	- Varnish, Moisture-and-Fungus-Resistant (For Treatment of Communications, Electronic, and Associated Equipment).
MIL-C-675	- Coating of Glass Optical Elements (Anti-Reflection).
MIL-T-704	- Treatment and Painting of Materiel.
MIL-S-3534	- Surveying Instruments and Accessories, Packaging of.
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-W-13518	- Wood Preservative: Tetrachlorophenol and Pentachlorophenol, Surface Sealing Compound.
MIL-R-14328	- Rubber Sheet: Synthetic, Medium Soft, General-Purpose Gasket Material (For Extreme Climatic Conditions).

STANDARDS

FEDERAL

FED. STD. No. 595	- Colors.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.

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MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other Publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS (NBS)

Special Publication 374 - Method for Determining the Resolving Power of Photographic Lenses.

(Application for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

SOUTHERN CALIFORNIA AIR POLLUTION CONTROL DISTRICT

Rule 102 - Photochemically Reactive Solvents.

(Application for copies should be addressed to the Southern California Air Pollution Control District, 434 South San Pedro Street, Los Angeles, CA 90013.)

3. REQUIREMENTS

3.1 Description. The theodolite shall be a direction-type, optical-reading instrument of first-order astronomic precision. The theodolite shall consist of a telescope assembly, base assembly, suspension level, illumination system, cases, and tripod.

3.2 First article (preproduction model). The contractor shall furnish one theodolite for examination and testing within the time frame specified (see 6.2) to prove prior to starting production that

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his production methods and choice of design detail will produce theodolites that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Initial production. When specified (see 6.2), the contractor shall furnish to the Government one or more theodolites for inspection as specified in 4.4.

3.4 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.6). Materials shall be corrosion resistant or treated to resist corrosion.

3.5 Environmental conditions.

3.5.1 Operating temperature. The theodolite shall operate as specified herein in any ambient temperature from plus 125° F to minus 50° F (plus 51.7° C to minus 45.6° C).

3.5.2 Storage temperature. The theodolite, tripod, cases, tools, and accessories shall not be damaged by storage under the following conditions:

Air temperature:

High - plus 155° F (plus 68.3° C) time period, 4 hours.
Low - minus 50° F (minus 45.6° C) time period, 6 hours.

3.5.3 Rainfall. The theodolite base and telescope assemblies and their transport/storage cases shall show no signs of water penetration when subjected to rainfall at a rate of 3 inches, plus or minus one inch (76.2 millimeters (mm), plus or minus 25.4 mm) per hour.

3.5.4 Sand and dust. When the theodolite and transport/storage cases are subjected to blowing sand and dust, the theodolite shall not be damaged, shall not show sand and dust penetration in the telescope or in the optical system, and shall not show lens abrasion.

3.5.5 Vibration. The theodolite base assembly and telescope assembly, when stored in their respective transport/storage cases, shall not be damaged by vibration when tested as specifiied in 4.6.2.15.

3.5.6 Shock. The theodolite base assembly and telescope assembly, when stored in their respective transport/storage cases, shall not be damaged by shock when tested as specified in 4.6.2.16.

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3.6 Telescope assembly. The telescope assembly shall consist of a telescope, optical system, vertical setting circle assembly, vertical circle assembly, and two Horrebow-Talcott-type levels. The telescope assembly shall have round polished surfaces on each end and shall rest on the telescope support bearings.

3.6.1 Telescope. The telescope, including an eyepiece assembly, shall be the "broken" or elbow type and and shall rotate about both the vertical and horizontal axes. The telescope tube shall be located at the center of the horizontal axis. The focal axis of the eyepiece shall remain horizontal for any tilt of the telescope. The tilt of the telescope shall be adjustable. The telescope shall permit the nighttime observation in clear weather of stars listed in astronomical ephemerides to the 7th magnitude. Open sights shall be mounted on the telescope. The telescope shall be equipped with a fine adjusting screw and a vertical clamp. The horizontal axis shall be adjustable from the trunnion.

3.6.1.1 Eyepiece assembly. The eyepiece assembly shall consist of an eyepiece and a micrometer mounted on the telescope. The eyepiece and the micrometer shall rotate together about the optical axis within limits of 90 degrees of arc as determined by two adjustable stops.

3.6.1.2 Tracking eyepiece micrometer. The telescope shall be provided with an eyepiece and tracking micrometer. A movable line shall always remain in the center of the field of view with the eyepiece mounted on a micrometer slide. Movement of the eyepiece and slide shall be precisely controlled by a measuring screw operable from outside the housing or by a pair of hand wheels to be used for continuous tracking. A micrometer drum divided into 100 equal parts shall be connected to the measuring screw. Each drum interval shall be equivalent to 1.5 seconds of arc. For each revolution of the drum, the movable line shall be displaced by an equal amount, not to exceed 2.5 minutes of arc. The range of the micrometer shall be 12 drum revolutions. Each revolution shall be indicated by a graduation on the fixed reticle. There shall be an inner contact drum which is attached to, and moves synchronously with, the micrometer drum. The contact drum shall be an agate disc into which 10 equally spaced platinum contact strips have been set. The contact strips shall be of sufficient width to close an electric circuit for 0.1 second of time at average tracking speed of an observed star. One contact strip shall be identified as the index or zero mark by the placement of two adjacent platinum strips, one on either side. A bracket mounted on the drum, shall close an electrical circuit as the contact strips rotate past, and send a signal to the chronograph via a direct cable. The bracket shall be connected to an exterior 3-prong plug outlet which will be used as a cable connection.

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3.6.1.3 Reticles, telescope. The reticles shall consist of a pattern of opaque lines permanently registered on optical glass and shall be enclosed in dust-resistant, moisture-resistant housings. Reticle lines shall be uniform in width and clean-cut and shall subtend not more than 3 seconds of arc nor less than 2.5 seconds of arc. The eyepiece micrometer reticle, which is movable, shall be registered with a single center line. The telescope reticle, which is fixed, shall be registered with 7 parallel lines, one in the center and 3 on each side, at distances from the center which subtend angles of 1, 3, and 6 minutes of arc respectively. A minimum of 10 graduations, numbered parallel to the 5 lines, and having 5 graduations on each side of the center, shall be registered. Each graduation shall indicate one full turn of the tracking eyepiece micrometer. Perpendicular to the 5 parallel lines, and on the same reticle, there shall be a single centerline, the top third of which shall be bifilar. At the fifth and fifteenth drum revolution a bifilar line shall be registered parallel to the centerline.

3.6.2 Optical system. The optical system shall produce a sharp image throughout the entire range of focus and shall be free from internal reflections, spherical and chromatic aberration, coma, astigmatism, and distortion. The field of view of the telescope shall be substantially flat with no marked curvature near the edge of the field. The principal axis of the optical system shall be aligned and coincident with the collimation of the telescope adjustment. Optical quality of the telescope shall be such that stars of 7th magnitude or brighter may be observed. Optical properties shall conform to the following requirements:

Magnification	
minimum	60X
maximum	80X
Clear objective aperture	2.8 inches (70 mm)
Diameter of field of view	46 minutes of arc
Resolution	The telescope shall clearly resolve Pattern 80 contained in National Bureau of Standards Special Publication 374 when the test pattern is at minimum focus distance from the telescope.

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3.6.2.1 Coated optics. Except for the registered faces of the reticles, the internal lens-air surfaces of the telescope shall be coated with a reflection-reducing film in accordance with MIL-C-675.

3.6.3 Vertical setting circle assembly. The vertical setting circle assembly shall consist of a vertical setting circle, an eyepiece, and a level. The setting circle shall be equipped with a drive for rapid setting of any tilt of the telescope to 1-minute precision. The eyepiece shall be located adjacent to the telescope eyepiece. Sensitivity of the level shall be not more than 1 minute of arc for each 2-mm movement of the bubble.

3.6.4 Vertical circle assembly. The vertical circle assembly shall consist of a vertical circle, circle drive, optical micrometer, micrometer drive, vertical clamp, and coincident-type level. The vertical circle shall be a 360-degree optical glass circle graduated in 4-minute intervals, with each degree numbered. The vertical circle shall be independently rotatable and may be set to any required angle. The optical micrometer shall have a graduation interval of 0.2 second. Sensitivity of the level shall not exceed 2 seconds of arc for each 2-mm movement of the bubble. Graduation error of the arithmetic mean of two diametric lines, including observation error (maximum deviation or t in Figure 1), shall not exceed $\pm$0.50 second. The tolerance of the arithmetic mean of two diametric lines (t in Figure 1) shall be $\pm$1.00 second.

3.6.5 Horrebow-Talcott-type levels. Two Horrebow-Talcott-type levels shall be mounted as a pair on and perpendicular to the horizontal axis. A clamping arrangement shall permit the measurement of changes in tilt of the telescope. The length of the bubble in each level shall be independently adjustable. Sensitivity of the levels shall be between 0.8 second and 1.5 seconds of arc for each 2-mm movement of the bubble.

3.7 Base assembly. The base assembly shall consist of a horizontal circle assembly, setting circle, horizontal and vertical slow motion screws, horizontal clamp, three leveling foot screws, trunnion, circular level, switches, and electrial connections. The vertical axis system shall be supported on conical ball bearings.

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3.7.1 Horizontal circle assembly. The horizontal circle assembly shall consist of a horizontal circle, circle drive, optical micrometer, and micrometer drive. The horizontal circle shall be a 360-degree optical glass circle graduated in 2-minute intervals. The horizontal circle shall be independently rotatable and may be set to any required angle. The optical micrometer shall have a graduation interval of 0.1 second. The graduation error of the arithmetic mean of the two diametrical lines, including the error of observation (maximum deviation or t in Figure 1), shall not exceed ±0.50 second of arc. The tolerance of the arithmetic mean of two diametrical lines (t in Figure 1), shall be 1.00 second with all eyepieces.

3.7.2 Circular level. A circular level shall be attached to the base. Sensitivity of the level shall be 4 minutes, plus or minus 1 minute.

3.8 Suspension level. A suspension level shall be suspended from the round polished surfaces on the ends of the telescope assembly. The bubble length shall be adjustable. Sensitivity of the level shall be not more than 1.2 seconds of arc for each 2-mm movement of the bubble. Graduations shall extend far enough to include the ends of the bubble at any operation ambient temperature. The factory-determined sensitivity of the level shall be legibly marked in the case supplied for the level.

3.9 Metal housings. Metal housings shall protect all screw threads, circles, and reticles from moisture and dust.

3.10 Illumination. The theodolite shall be illuminated by electrical means. Electrical illumination shall be supplied by one 12 V NiCd battery. Lighting attachments shall illuminate the telescope, horizontal and vertical circles, and the vertical setting circle with a separate switch for each unit. Lighting attachments shall include frosted glass covers or 12 V frosted lamps. An additional 12 V NiCd battery shall be provided when specified (see 6.2).

3.10.1 Reticle illumination. Brilliance of the reticle illumination shall be adjustable. Illumination of the reticle shall not affect the optical properties of the telescope.

3.11 Wiring. The theodolite shall be wired for illumination. External wiring shall be extra flexible. Connecting cords from

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the theodolite to the battery case and the chronograph shall have plugs to fit into terminal connections at the base of the theodolite. Connecting cords to the battery case shall be 6 feet (1.8 m) long. External wiring shall not crack or show evidence of damage when subjected to the operating temperatures specified herein and when tested as specified in 4.6.2.6.

3.12 Transport and storage cases. Separate cases shall be supplied for the base assembly, telescope assembly, suspension level, battery, and tools and accessories. All corners and edges of each case shall be reinforced. Each case shall have a lock or latch and carrying handles. Each case shall have cushioning material to prevent damage from shock and vibration during transit or storage. Cases for the base assembly and telescope assembly shall have a dessicant container in the bottom. Material of construction which meets commercially acceptable standards is at the option of the contractor. Cases for the base assembly and telescope assembly shall serve for both transport and storage.

3.12.1 Battery case. The battery case shall accommodate one 12V NiCd battery. There shall be two exterior receptacles; one for connecting the battery to the theodolite, and one for connecting the handlamp. The case shall be provided with a dimming rheostat and switch, and shall accommodate the lighting accessories specified in 3.12.1.1.

3.12.1.1 Lighting accessories. The following accessories shall be furnished in the quantities specified herein:

Item	Quantity
Connector cord for the base of the theodolite	1
Connector cord for recording micrometer	1
Headlamp	1
Lamps, electric	4

3.12.2 Tool and accessories case. A case of the size and shape necessary to contain tools and accessories to maintain and adjust the theodolite shall be supplied. The case shall accommodate tools and accessories listed in 3.12.2.1.

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3.12.2.1 Tools and accessories. The following tools and accessories shall be provided:

Item	Quantity
Brushes	2
Cleaning sticks	3
Chamois skin	1
Jewelers screwdrivers	3
Large adjusting pins	2
Fork and pin wrench	1
Tweezers (pair)	1
Common screwdrivers	3
Connecting cable with switch and resistor	1
Emergency lamp with cable	1
Handlamp with cable	1
Connecting cable for chronograph	1
Connecting cables for instrument	2
Grease can	1
Small adjusting pins	4
Spare bulbs	8

3.12.3 Electric lamps. Lamps shall be 12V.

3.12.4 Electric lamp sockets. Electric lamp sockets shall be a size and type to fit the lamps specified in 3.12.3 except for the autocollimating eyepiece illumination which may be at the contractor's option.

3.13 Tripod. Unless otherwise specified (see 6.2), a tripod shall be supplied. The tripod shall be a rigid, nonextension-leg type. The legs shall be hinged to a headplate. The hinges and the headplate shall be nonmagnetic metal or alloy. The hinges shall be adjustable for tension. The legs shall be seated in metal shoes. The legs shall be seasoned hardwood such as white ash, yellow birch, European birch, hickory, pecan, or hard maple. Slope of the grain of the wood shall be not more than 1 inch in 15 inches. The wood shall be free from knots. A leather or webbing binding strap shall be attached to one of the

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legs near the shoes for holding the legs together when not in use. The tripod shall weigh not more than 40 pounds (18 kilograms). Unless otherwise specified (see 6.2), a leather cap for the head of the tripod shall be supplied.

3.14 Theodolite cover. Unless otherwise specified (see 6.2), a light-weight, flexible, waterproof cover for use when the theodolite is assembled shall be supplied.

3.14.1 Winterization kit. When specified (see 6.2) a winterization kit shall be furnished with the theodolite. The winterization kit shall be designed primarily to assure satisfactory operation of the theodolite, and protect the operator at extremely low operational temperatures. With the winterization attachments in place, the theodolite shall fit readily into its carrying cases. The kit shall consist of a canvas carrying case containing rubber knob covers and eyepiece shields.

3.14.1.1 Rubber knob covers. Rubber knob covers shall be provided for all control knobs, leveling screws, and levers. The knob covers shall be a shape and size to permit manipulation of the controls by an operator wearing heavy mittens. The knob covers shall be rubber conforming to MIL-R-14328 and shall fit snugly.

3.14.1.2 Eyepiece shields. Eyepiece shields shall be provided for the telescope eyepiece and microscope eyepieces. The shields shall be hard rubber or laminated fabric material. They shall be readily attached or removed without cracking or splitting at any temperature to minus 50° F (minus 45.6° C). The shields shall fit snugly over the eyepieces without disturbing the focus or decreasing the field of view of the telescope or reading microscopes. The shields shall provide protection against frosting of the exterior surfaces of the eyepiece lenses without interfering with the accuracy of the instrument. Two sets of eyepiece shields shall be furnished.

3.15 Lubrication. Unless otherwise specified (see 6.2), the theodolite shall be lubricated with grease conforming to MIL-G-10924.

3.16 Fungus and moisture resistance. The electrical circuitry, including all components and connections, except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, Composition I or II, as applicable (see 3.16.1), with 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish.

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(a) Components or circuit elements that are inherently fungus- and moisture-resistant or which are hermetically sealed need not be treated.
(b) Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.16.1 Composition. Composition II shall be used only in the case when local air pollution regulations governing the application of varnish precludes the use of Composition I. When Composition II is used, the contractor shall provide evidence to the Government that the use of Composition II is required, and shall certify that the Composition II material complies with Rule 102, Southern California Air Pollution District.

3.17 Identification marking. The theodolite shall be identified in accordance with MIL-STD-130.

3.18 Treatment and painting. The portions of the theodolite, tripod, and cases normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704. The color of theodolite shall conform to FED. STD. No. 595, Color No. 24159. The wooden part of the tripod legs shall not be colored and the grain of the wood shall be visible. The wood shall be treated with a wood preservative surface sealer conforming to MIL-W-13518, Type I. The wood shall be finished with varnish conforming to TT-V-121.

3.19 Workmanship. The theodolite shall show precision workmanship. Movable parts shall neither bind nor move loosely throughout their range of movement. Bearing surfaces shall show no scratches or surface blemishes. Components or assemblies shall show no evidence of forced attachment. Materials shall be free from all defects and imperfections that may affect the operation or life of the theodolite or components.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Initial production inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).
(d) Inspection comparison (see 4.7).
(e) Inspection of packaging (see 4.8).

4.3 Preproduction inspection.

4.3.1 Examination. The preproduction theodolite shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The preproduction theodolite shall be tested as specified in 4.6.2. Failure of any test shall be cause for rejection.

4.4 Initial production inspection. When specified (see 3.3), one or more initial production theodolites will be selected at random by the Government from the theodolites being produced by production tooling and will be examined as specified in 4.6.1 and tested as specified in 4.6.2, to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of an initial production theodolite shall not exclude the remaining theodolites from the quality conformance inspection and acceptance provisions specified in Section 4. In

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addition to any test specified as part of the initial production test, the Government reserves the right to conduct any or all other tests contained in this specification as part of the initial production test, and failure of such additional tests shall have the same effect as failure of those tests specified as initial production tests.

4.4.1 Inspection failure. Failure of an initial production theodolite to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the initial production theodolite and shall be cause for refusal by the Government to continue acceptance of production theodolites until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on theodolites previously accepted and produced under the contract. Any deficiencies found as a result of the initial production inspection will be considered prima facie evidence that all theodolites accepted prior to the completion of initial production inspection are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Sampling. Sampling for tests shall be in accordance with MIL-STD-105.

4.5.2 Examination.

4.5.2.1 Individual. Each theodolite shall be examined for the defects listed in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.5.3 Tests.

4.5.3.1 Individual. Each theodolite shall be tested as specified in 4.6.2.17 and 4.6.2.18. Failure of any test shall be cause for rejection.

4.5.3.2 Samples. Samples selected in accordance with 4.5.1 shall be tested as specified in 4.6.2.7 through 4.6.2.16. AQL shall be 2.5 percent defective.

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4.6 Inspection procedure.

4.6.1 Examination. The theodolite shall be examined as specified herein for the following defects.

101. Materials not as specified.
102. Telescope assembly not as specified.
103. Base assembly not as specified.
104. Suspension level not as specified.
105. Factory-determined sensitivity of the suspension level not legibly marked in the suspension level case.
106. Metal housings for all screw threads, circles, and reticles not provided.
107. Illumination not as specified.
108. Cases not as specified.
109. Lighting accessories missing or not as specified.
110. Tools and accessories not as specified.
111. Electric lamps and lamp sockets not as specified.
112. Winterization kit or components not as specified.
113. Tripod not as specified.
114. Theodolite cover not as specified.
115. Lubrication not as specified.
116. Fungus and moisture resistance not as specified.
117. Identification marking missing, incomplete, or illegible.
118. Treatment and painting not as specified.
119. Workmanship not as specified.

4.6.2 Tests.

4.6.2.1 Operational. The operational test shall be conducted when specified herein. The theodolite shall be operated and inspected as follows to determine its operational capabilities:

Manipulate all movable parts and inspect	- For freedom of movement and non-interference with other parts in various operating positions throughout their maximum ranges.
Fit or attach all accessories to the theodolite and inspect	- For functionality as specified herein.

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Inspect the levels	- For free movement to determine if the bubble remains within the graduations and if the bubble ends of the coincidence level remain visible.
Rotate the theodolite about its axes, and inspect the illuminating system	- For uniform, steady operation.
Inspect microscope focus	- For uniform and clear focus around each entire circle.
Inspect parts, components, accessories	- For damage.

4.6.2.2 Low temperature operation. The assembled theodolite mounted on the tripod, with the cases, tools, and accessories, shall be placed in a temperature-control chamber. The temperature shall be gradually lowered to minus 50° F (minus 45.6° C) and maintained at that temperature until the theodolite is in thermal equilibrium. While at this temperature, the theodolite shall then be connected to the battery, which is not subject to this test. Each switch shall be turned on and each light shall be checked for operation. The telescope assembly shall be rotated about its horizontal axis as far as the design of the theodolite will permit. The telescope assembly support shall be rotated about its vertical axis 360 degrees. All operating devices pertinent to the operation of the theodolite shall be moved or positioned to various operating positions. Evidence of any operating device requiring force to move, or malfunction of the illumination system shall constitute failure of this test.

4.6.2.3 Low temperature storage. After successful completion of the low temperature operation test, and maintaining a temperature of minus 50° F (minus 45.6° C), disconnect the battery from the theodolite and remove the battery from the control chamber. The temperature in the control chamber shall then be lowered to minus 65° F (minus 53.9° C) and maintained at that temperature for 3 days. The temperature shall then be raised to plus 79° F (plus 26.1° C) over a period of not less than 10 hours. The battery shall then be connected to the theodolite. The illumination system and each operating device shall again be checked for operation. Evidence of damage to the illumination system, any operating device or functional part of the theodolite, tripod, cases, or tools and accessories shall constitute failure of this test.

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4.6.2.4 High temperature operation. The assembled theodolite mounted on the tripod, with the cases, tools, and accessories, shall be placed in a temperature-control chamber. The temperature shall be gradually raised to plus 125° F (plus 51.7° C) and maintained at that temperature until the theodolite is in thermal equilibrium. While at this temperature, the theodolite shall then be connected to the battery, which is not subject to this test. Each switch shall be turned on and each light shall be checked for operation. The telescope assembly shall be rotated about its horizontal axis as far as the design of the theodolite will permit. The telescope assembly support shall be rotated about its vertical axis 360 degrees. All operating devices pertinent to the operation of the theodolite shall be moved or positioned to various operating positions. Evidence of any operating device requiring force to move, or malfunction of the illumination system or cracking, rupturing of the level vials, shall constitute failure of this test.

4.6.2.5 High temperature storage. After successful completion of the high temperature operation test and maintaining a temperature of plus 125° F (plus 51.7° C), disconnect the battery from the theodolite. The control chamber temperature shall be increased to plus 155° F (plus 68.3° C) and maintained at that temperature for 4 hours. The temperature shall then be lowered to plus 79° F (plus 26.1° C) over a period of not less than 4 hours. The battery shall then be connected to the theodolite. The illumination system and each operating device shall again be tested for operation. Evidence of damage to the illumination system, any operating device, tripod, cases, or tools and accessories shall constitute failure of this test.

4.6.2.6 Temperature tests of the wiring. Temperature tests of the wiring shall be conducted by using the battery and chronograph cords. Temperature tests of the wiring may be conducted concurrent with the temperature tests of the theodolite.

4.6.2.6.1 Low temperature. The battery and chronograph cords shall be placed in a temperature-control chamber in a nonflexed position. The temperature shall be gradually lowered to minus 40° F (minus 40° C) and maintained at that temperature for 4 hours. Without removal from the chamber, the two cords shall be bent around a 0.375-inch (9.5 mm) diameter mandrel at a rate of 12 turns per minute. The cords shall then be removed from the chamber and examined for cracks or flaws under a glass having a magnification of 3 diameters. Evidence of damage to the cords shall constitute failure of this test. The cords shall then be returned to the chamber and subjected to the storage temperature of

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minus 50° F (minus 45.6° C) for a period of 3 days. When the temperature has been raised to plus 79° F (plus 26.1° C), the cords shall be examined again for cracks or flaws. Nonconformance to 3.11 shall constitute failure of this test.

4.6.2.6.2 High temperature. The battery and chronograph cords shall be placed in a temperature-control chamber in a nonflexed position. The temperature shall be gradually raised to plus 125° F (plus 51.7° C) and maintained at that temperature for 4 hours. Without removal from the chamber, the two cords shall be bent around a 0.375-inch (9.5 mm) diameter mandrel at a rate of 12 turns per minute. The cords shall then be removed from the chamber and examined for cracks or flaws under a glass having a magnification of 3 diameters. Evidence of damage to the cords shall constitute failure of this test. The cords shall then be returned to the chamber and subjected to the storage temperature of plus 155° F (plus 68.3° C) for a period of 4 hours. When the temperature has been lowered to plus 79° F (plus 26.1° C), the cords shall be examined again for cracks or flaws. Nonconformance to 3.11 shall constitute failure of this test.

4.6.2.7 Level sensitivity. Sensitivity of level vials shall be determined by mounting the level vials on a level trier or by any other method yielding values accurate within 5 percent. The plane of the graduated vials shall be changed the entire amount and the bubble driven over the whole vial step-by-step; first in one direction, then the other. The plane of the ungraduated vials shall be changed a sufficient amount to cause a change of not less than 4 mm in the position of the level bubble as indicated on a scale placed adjacent to the vials. Nonconformance to values specified in 3.6.3, 3.6.4, 3.6.5, and 3.8 shall constitute failure of this test.

4.6.2.8 Magnification. Magnifying power, in diameters, of the telescope shall be measured by focusing the telescope for parallel light with the aid of a low-power telescope placed behind the eyepiece of the telescope under test or by any other method yielding values accurate within 2 percent. Without change of focus, a transparent graduated scale shall be placed in front of the objective lens and as near thereto as practicable. The low-power telescope shall be replaced with a measuring microscope and focused on the image of the scale. Magnification of the telescope is then determined in diameters as the ratio of a given length of the scale to the length of its image in the exit pupil. Nonconformance to 3.6.2 shall constitute failure of this test.

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4.6.2.9 Effective aperture. A transparent scale graduated in millimeters shall be placed in contact with the front of the objective lens. The effective aperture is the visible portion of the scale when sighted through the eyepiece. Nonconformance to 3.6.2 shall constitute failure of this test.

4.6.2.10 Field of view. The horizontal width of the field of view shall be measured by the use of two targets. Place the two targets 1,640.5 feet (500 m) from the telescope. One target shall be on a horizontal crossline at the extreme left of the field of view, and the second shall be on a horizontal crossline at the extreme right of the field of view. Measure the angle between the two targets. The vertical width of the field of view shall be measured in the same manner except that the two targets shall be on a vertical crossline. Nonconformance to 3.6.2 shall constitute failure of this test.

4.6.2.11 Minimum focus. The telescope shall be set at minimum focus. A graduated scale or object marked with clearly defined lines shall be moved along the line of sight from the telescope until the markings are clearly visible by the lens of the telescope. The shortest distance from the telescope to a point on the line of sight at which the markings become clearly visible is the minimum focus value. Nonconformance to 3.6.2 shall constitute failure of this test.

4.6.2.12 Resolution. Observations for resolution shall be made using National Bureau of Standards Special Publication 374, Pattern 80. The pattern shall be placed 100 feet (30.5 m) from the telescope. Nonconformance to 3.6.2 shall constitute failure of this test.

4.6.2.13 Rainfall. With the telescope and base assemblies of the theodolite stored in their respective transport/carrying cases, the cases shall be subjected to a rainfall test. The test facility shall be not less than 36 inches (0.91 meter) high, 36 inches (0.91 meter) wide, and 36 inches (0.91 meter) deep. The facility shall be equipped with water spray nozzles to provide a uniform flow of water over the test area at a rate of 3 inches plus or minus 1 inch (0.08 meter plus or minus 0.02 meter) per hour. The direction of the rainfall shall be variable up to 45 degrees from the vertical. Water temperature shall be 52° F to 68° F (11.1° C to 20° C). The theodolite, in the case, shall be subjected to this test for a period of not less than 1 hour. Presence of water in the transport/carrying case, the telescope or optical system shall constitute failure of this test.

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4.6.2.14 Sand and dust. With the telescope and base assemblies of the theodolite stored in their respective transport/carrying cases, the cases shall be subjected to a sand and dust test. The test facility shall be not less than 36 inches (0.91 meter) high, 36 inches (0.91 meter) wide, and 36 inches (0.91 meter) deep. The sand and dust (140-mesh silica flour) shall pass through the chamber at a velocity of 100 to 500 feet per minute (30.5 to 152.4 meters per minute). The density of the sand and dust within the test facility shall be maintained at 0.1 to 0.5 grams per cubic foot (3.5 to 17.7 grams per cubic meter). The internal temperature shall be maintained at 77° F plus or minus 5° F (25° C plus or minus 2.8° C). The relative humidity shall not exceed 30 percent. The test duration shall be not less than 15 minutes. The presence of sand or dust in the telescope, in the optical system, or on the bearings, or the presence of lens abrasion shall constitute failure of this test.

4.6.2.15 Vibration. With the telescope and base assemblies stored in their respective transport/carrying cases, the cases shall be subjected to vibration testing in accordance with MIL-STD-810, Method 514.2, Procedure X, Table 514.2-VII, Figure 514.2-7, Curve AW, except that only 5 sweeps of vibration cycling shall be performed for a total of 1 hour of vibration in each axis. Upon completion of the test, the theodolite shall be assembled and tested as specified in 4.6.2.1. Evidence of damage to the theodolite or to the cases shall constitute failure of this test.

4.6.2.16 Shock. With the telescope and base assemblies stored in their respective transport/carrying cases, the cases shall be dropped in an upright position from a height of 6 inches (152 mm) onto a level wood floor. The theodolite shall then be assembled and subjected to the operational test specified in 4.6.2.1. Evidence of damage to the theodolite or to the cases shall constitute failure of this test.

4.6.2.17 Horizontal circle. This test shall be reported on a test report form similar to that shown in Figure 1. A duplicate copy of the completed test report shall be placed within the carrying case of the theodolite to which the report applies. The test shall be conducted in accordance with the following instructions which shall be printed on the reverse side of the test report form:

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THEODOLITE CIRCLE TEST INSTRUCTIONS

This form is based on a series of observations on two collimators 36 degrees apart and on a horizontal plane having an elevation approximately the same as that of the horizontal axis of the instrument being tested.

Coincidence of the micrometer shall be made twice for each collimator pointing. The two readings are entered in columns I and II. Recording of minutes is not necessary but may be made in column M if desired. The mean of the two readings is entered in column III. Column IV, Angle 1, is the difference between two adjacent means.

Series I is made up of two sets of angles:
The 1st set consists of 5 angles between zero and 180 degrees.
The 2nd set consists of 5 angles between 180 and 360 degrees.

Means and angles are determined as above, and column III, Angle 2, for the second set is transcribed into column V, opposite the corresponding angles of the first set. Means of the two angles are determined and entered in column VI, Mean. These are summed, and a mean is again determined. Residuals are found and entered in column VII, V.

Zero is entered opposite the first residual, and the algebraic sum of the residuals is entered in column VIII. The last value in this column should be the last value of column VII, but with opposite sign. These values are added algebraically, and a mean is determined. This mean is subtracted algebraically from the values of column VIII to give column IX, (t). The summation of the column should equal zero. Column X, Δ, is the algebraic difference of column IV, Angle 1 and column V, Angle 2.

Series II is taken in the same manner, except that the "settings" or "positions" of the circle are halfway between the "settings" or "positions" of series I.

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The Computations are made by transcribing (t), V, and Δ from series I and II, raising to the square as indicated, and solving the stated formulae. The tolerance for (t) is plus and minus 0.50 second. The maximum plus or minus deviations are obtained by scanning the column (t) in the Computations. Neither value shall exceed 1.00 second; $(t)_m$ is the root mean square value of (t). There is no fixed value for this root, but a small value indicates high accuracy of the circle.

Evidence of any horizontal circle whose accuracy exceeds the tolerances for (t) and the maximum deviations specified herein shall constitute failure of this test.

4.6.2.18 Vertical circle. This test shall be reported on a test report form similar to that shown in Figure 1. A duplicate copy of the completed test report shall be placed within the carrying case of the theodolite to which the report applies. The test shall be conducted in accordance with the following instructions which shall be printed on the reverse side of the test report form:

THEODOLITE CIRCLE TEST INSTRUCTIONS

This form is based on a series of observations on two collimators 36 degrees apart and on a vertical plane.

Coincidence of the micrometer shall be made twice for each collimator pointing. The two readings are entered in columns I and II. Recording of minutes is not necessary but may be made in column M if desired. The mean of the two readings is entered in column III. Column IV, Angle 1, is the difference between two adjacent means.

Series I is made up of two sets of angles:
The 1st set consists of 5 angles between zero and 180 degrees.
The 2nd set consists of 5 angles between 180 and 360 degrees.

Means and angles are determined as above, and column III, Angle 2, for the second set is transcribed into column V, opposite the corresponding angles of the first set. Means of the two angles are determined and entered in column VI, Mean. These are summed, and a mean is again determined. Residuals are found and entered in column VII, V.

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Zero is entered opposite the first residual, and the algebraic sum of the residuals is entered in column VIII. The last value in this column should be the last value of column VII, but with opposite sign. These values are added algebraically, and a mean is determined. This mean is subtracted algebraically from the values of column VIII to give column IX, (t). The summation of the column should equal zero. Column X, Δ , is the algebraic difference of column IV, Angle 1, and column V, Angle 2.

Series II is taken in the same manner, except that the "settings" or "positions" of the circle are half-way between the "settings" or "positions" of series I.

The Computations are made by transcribing (t), V, and Δ from series I and II, raising to the square as indicated, and solving the stated formulae. The tolerance for (t) is plus or minus 0.50 second. The maximum plus or minus deviations are obtained by scanning the column (t) in the Computations. Neither value shall exceed 1.00 second; $(t)_m$ is the root mean square value of (t). There is no fixed value for this root, but a small value indicates high accuracy of the circle.

Evidence of any vertical circle whose accuracy exceeds the tolerances for (t) and the maximum deviations specified herein shall constitute failure of this test.

4.7 Inspection comparison. The Government may select theodolites at any time during the contract production period and subject these theodolites to the examination and tests specified in 4.4 to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government, on theodolites selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction model and initial production theodolites. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison, and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.7.1 Inspection failure. Failure of an inspection comparison theodolite to meet any requirement specified herein during and as a result of the examination and tests specified in 4.7 shall be cause for

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rejection of the inspection comparison theodolite and shall be cause for refusal by the Government to continue acceptance of production theodolites until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on theodolites previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all theodolites accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.8 Inspection of packaging. Level A or Level B preservation, packing and marking shall be examined and tested in accordance with the quality assurance provisions of MIL-S-3534. Commercial preservation, packing and marking shall be examined for conformance to MIL-STD-1188.

5. PACKAGING

5.1 Preservation, packing and marking. Preservation, packing, and marking shall be Level A, Level B, or Commercial, as specified (see 6.2).

5.1.1 Level A or Level B. Level A or Level B preservation, packing, and marking shall be in accordance with MIL-S-3534.

5.1.2 Commercial. Commercial preservation, packing, and marking shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The theodolite with accessories is intended for astronomic observations for latitude, longitude, time and azimuth which must meet first order surveying specifications in Department of Army TM5-442, Precise Astronomical Surveys.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of preproduction model (see 3.2).
(c) When initial inspection is required and number of theodolites to be furnished (see 3.3).
(d) When an extra 12 V NiCd battery is required (see 3.10).

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(e) When tripod shall not be furnished (see 3.13).
(f) When leather cap shall not be furnished (see 3.13).
(g) When waterproof cover shall not be furnished (see 3.14).
(h) When winterization kit shall be furnished (see 3.14.1).
(i) When the theodolite shall not be lubricated as specified (see 3.15).
(j) Degree of preservation and degree of packing required (see 5.1).

6.3 Preproduction model. Any changes or deviations of production theodolites from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the contractor of his obligation to furnish theodolites conforming to this specification.

6.4 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each theodolite.

6.5 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools, as necessary (including any special tools), and instructions regarding shipment of theodolites.

6.6 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.4).

Custodian:
Army - ME

Preparing activity:
Army - ME

Project 6675-0385

Review activity:
DLA - GS

User activity:
Navy - YD

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THEODOLITE CIRCLE TEST

OBSERVER
RECORDER
CHECKED BY
DATE
TYPE
INS. NO
CIRCLE NO

SERIES I
1ST SET

COLUMN	I	II		III	IV	V	VI		VII	VIII	IX	X
POINTING			M		∠1	∠2	MEAN	RES	V		(i)	Δ
DEG												
0								1				
36												
36								3				
72												
72								5				
108												
108								7				
144												
144								9				
180												
2ND SET					∠2	$\frac{\Sigma}{\Sigma+5}$			$\frac{\Sigma}{\Sigma\div5}$		$\Sigma\cdot0$	
180												
216												
216												
252												
252												
288												
288												
324												
324												
0												

SERIES II
1ST SET

DEG					∠1
18					
54					
54					
90					
90					
126					
126					
162					
162					
198					

COMPUTATIONS

RES	(i)	$(i)^2$	V	VV	Δ	ΔΔ
1						
2						
3						
4						
5						
6						
7						
8						
9						
10						

$M^2_1 = \frac{\Sigma VV}{9}$ $\quad (i)^2_m = \frac{\Sigma (i)^2}{10}$

$M^2_2 = \frac{\Sigma \Delta\Delta}{40}$ $\quad (i)_m = \pm$

$2r^2 = M^2_1 - M^2_2$

$r = \pm$ $\quad$ MAX. DEVIATIONS + −

2ND SET					∠2	∠1	MEAN	RES	V		(i)	Δ
198								2				
234												
234								4				
270												
270								6				
306												
306								8				
342												
342								10				
18												

FIGURE 1. Theodolite circle test.

X-3600

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DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-T-52115B Theodolite, Surveying: Directional, Astronomical, 0.1 Second, with Accessories, Tripod, and Cases

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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